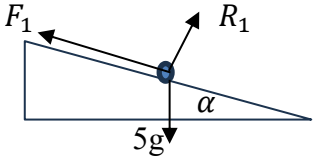
Question			Answer	Marks	AO	Guidance
14	(a)		$[-5g\mathbf{j} + R\mathbf{j} \pm F\mathbf{i} + 9\mathbf{i} + 20\mathbf{j} = \mathbf{0}]$			Allow wrong or missing vector equation
			Horizontal component $F = 9$	**B1**	**3.1b**	Soi. Allow $\boldsymbol{F} = -9\mathbf{i}$ or $F = -9$
			Vertical direction $R - 5g + 20 = 0$	**M1**	**3.1b**	Soi Vertical equation – weight and 20 involved and not 9. Allow sign errors
			$R = 5g - 20$	**A1**	**1.1**	Correct equation soi
			On the point of sliding $F = \mu R$ so	**M1**	**3.4**	Uses their F and R to evaluate μ. Where $F = \mu \times 5g$ is seen it must be explicit that $5g$ is their reaction and not weight
			So $\mu = \frac{9}{29} = [0.310$ to 3 sf$]$	**A1FT**	**1.1**	FT their R and F. Final answer must follow from their working and be positive Ignore unsupported $\mu = 0.310$ seen.
				[5]		

Question			Answer	Marks	AO	Guidance
14	(b)					Condone repeated use of F and R throughout even though their values have changed
			Resolve perpendicular $R_1 = 5g\cos\alpha$	**B1**	**3.4**	Allow $R - 5g\cos\alpha = 0$ or $F = \mu \times 5g\cos\alpha$ seen explicitly
			Resolve parallel $F_1 = 5g\sin\alpha$	**B1**	**3.3**	Allow for $5g\sin\alpha = \mu \times 5g\cos\alpha$ Allow sin/cos interchange if consistent with their R_1
			$F_1 = \frac{9}{29}R_1$ so $\tan\alpha = \mu \left[= \frac{9}{29}\right]$	**M1**	**1.1**	Allow for fraction which simplifies to 0.310 or better. FT their μ or 0.310 May be a quoted formula following B0B0
			$\tan^{-1}\frac{9}{29} = 17.2°$	**A1**	**2.1**	**AG** must follow $\mu = 0.310$ correctly calculated in (a)
				[4]		
14	(c)		N2L up the slope $-(5g\sin 17.2° + \frac{9}{29}5g\cos 17.2°) = 5a$	**B1** **B1FT** **M1**	**3.3** **3.4** **3.1b**	Correct weight term seen (Allow ± 14.5 oe) Friction term seen FT their μ (Allow ± 14.5 oe) Formulate equation of motion with their weight. Allow wrong or missing friction. Allow sign errors No extra forces.
			$a = -5.80$ ...	**A1**	**1.1**	Soi Allow 5.8 if down the slope used as positive direction clear (eg on diagram, or $u = -5$ or $a = -5.8$ used subsequently)
			$v = u + at$ with $v = 0, u = 5$ and their a	**M1**	**3.4**	$suvat$ equation(s) leading to a value for t. Allow sign errors
			$t = 0.862$ s	**A1FT**	**1.1**	FT their acceleration $\neq \pm g$ Using $\alpha = \tan^{-1}\frac{9}{29}$ gives 0.86067
				[6]		